Control, Complement Size, and Overt Controlees in Mandarin: An Experimental Study

Sam QU
samzqu.ling@gmail.com

Nick HUANG
znhuang@nus.edu.sg

15th International Workshop on Theoretical East Asian Linguistics(TEAL15)

June 2026, The Chinese University of Hong Kong

Background: PC vs EC

- Since Landau (2000), obligatory control has been observed to comprise two subtypes: Partial Control (PC) and Exhaustive Control (EC).

- The semantics of PC and EC differs in two important respects.

Background: PC vs EC

- Since Landau (2000), obligatory control has been observed to comprise two subtypes: Partial Control (PC) and Exhaustive Control (EC).

1. PRO exhaustivity.

'Gather' requires semantically plural subjects:

- A crowd **gathered** outside the building. ✓
- A student **gathered** outside the building. ✗

Background: PC vs EC

- Since Landau (2000), obligatory control has been observed to comprise two subtypes: Partial Control (PC) and Exhaustive Control (EC).

  1. PRO exhaustivity.

  - The chair$_i$ managed [PRO$_{i+j}$ to gather at 6]. ✗
  - The chair$_i$ managed [PRO$_i$ to arrive at 6]. ✓

  - The chair$_i$ preferred [PRO$_{i+j}$ to gather at 6]. ✓

from Landau (2000:14)

Background: PC vs EC

- Since Landau (2000), obligatory control has been observed to comprise two subtypes: Partial Control (PC) and Exhaustive Control (EC).

1. PRO exhaustivity.

EC predicates require PRO to be referentially **identical** to the controller:

- The chair$_i$ managed [PRO$_{i+j}$ to gather at 6]. ✗
- The chair$_i$ managed [PRO$_i$ to arrive at 6]. ✓

whereas PC predicates allow PRO to semantically include the controller:

- The chair$_i$ preferred [PRO$_{i+j}$ to gather at 6]. ✓

from Landau (2000:14)

Background: PC vs EC

- Since Landau (2000), obligatory control has been observed to comprise two subtypes: Partial Control (PC) and Exhaustive Control (EC).

2. Temporal exhaustivity.

PC verbs permit the complement event to be non-simultaneous with the matrix event,

- Yesterday, John wanted [PRO to solve the problem (tomorrow).] ✓

but most EC verbs forbid this:

- Yesterday, John managed [PRO to solve the problem (*tomorrow).] ✗

from Landau (2000:15)

Background: PC vs EC

- Parallel patterns of PC and EC have also been documented for **Mandarin Chinese** (Grano, 2015; He, 2024; Wurmbrand & Lohninger, 2023; N.Huang, 2018; C.-T. J. Huang, 2022, among others).

PRO: collective predicate diagnostics

- EC: ✗ 小明$_i$ 尝试 [PRO$_{i+j}$ (*一起) 开车 去 香港]
  xiaoming try together drive go Hong Kong
  "Xiaoming tries to drive to Hong Kong together."
- PC: ✓ 小明$_i$ 计划 [PRO$_{i+j}$ (一起) 开车 去 香港]
  xiaoming plan together drive go Hong Kong
  "Xiaoming plans to drive to Hong Kong together."

Background: PC vs EC

- Parallel patterns of PC and EC have also been documented for **Mandarin Chinese** (Grano, 2015; He, 2024; Wurmbrand & Lohninger, 2023; N.Huang, 2018; C.-T. J. Huang, 2022, among others).

PRO: collective predicate diagnostics

- EC: ✗ 小明$_i$ 尝试 [PRO$_{i+j}$ (*一起) 开车 去 香港]
  xiaoming try together drive go Hong Kong
  "Xiaoming tries to drive to Hong Kong together."
- PC: ✓ 小明$_i$ 计划 [PRO$_{i+j}$ (一起) 开车 去 香港]
  xiaoming plan together drive go Hong Kong
  "Xiaoming plans to drive to Hong Kong together."

Temporal: time mismatch diagnostics

- EC: ✗ 昨天 约翰 尝试 [PRO (*明天) 离开 北京]
  yesterday John try tomorrow leave Beijing
  "Yesterday, John tried to leave Beijing tomorrow."
- PC: ✓ 昨天 约翰 打算 [PRO (明天) 离开 北京]
  yesterday John plan tomorrow leave Beijing
  "Yesterday, John planned to leave Beijing tomorrow."

Background: PC vs EC

- In recent years, the PC-EC semantic differences have been found to have syntactic reflexes.
- Notably, PC and EC predicates have different C-selection (subcategorization) properties.

Background: PC vs EC

- Cross-linguistic evidence has shown that complement clauses can be semantically classified into three groups, based on their temporal and subject references (Wurmbrand & Lohninger, 2023):

	Proposition	**Situation**	**Event**
Type of Control	Independent, PC possible	PC possible	EC
Temporal Interpretation	Independent (speaker oriented)	Dependent, mostly future irrealis.	Tenseless, simultaneous

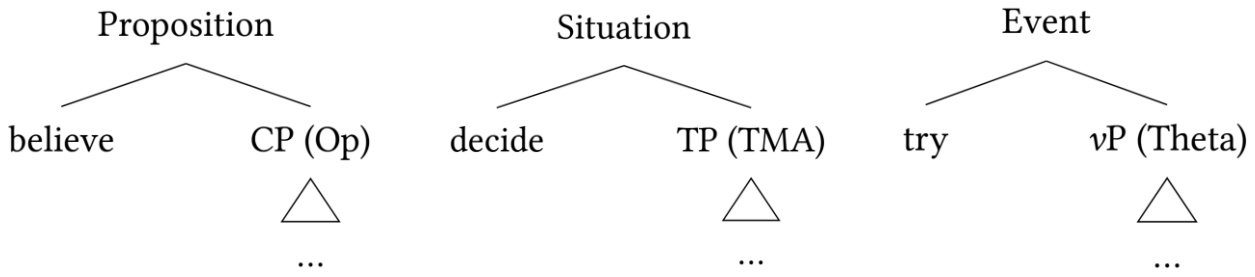
Background: PC vs EC

- Cross-linguistic evidence has shown that complement clauses can be semantically classified into three groups, based on their temporal and subject references (Wurmbrand & Lohninger, 2023):

	Proposition	**Situation**	**Event**
Type of Control	Independent, PC possible	PC possible	EC
Temporal Interpretation	Independent (speaker oriented)	Dependent, mostly future irrealis.	Tenseless, simultaneous

- The three semantic types are, **by default**, structurally realized as different syntactic categories:

Proposition	**Situation (PC)**	**Event (EC)**
Operator domain	TMA domain	Theta domain
Finite CP	TP	vP



Background: PC vs EC

- Cross-linguistic evidence has shown that complement clauses can be semantically classified into three groups, based on their temporal and subject references (Wurmbrand & Lohninger, 2023):

	Proposition	**Situation**	**Event**
Type of Control	Independent, PC possible	PC possible	EC
Temporal Interpretation	Independent (speaker oriented)	Dependent, mostly future irrealis.	Tenseless, simultaneous

- The three semantic types are, **by default**, structurally realized as different syntactic categories:

Proposition	**Situation (PC)**	**Event (EC)**
Operator domain	TMA domain	Theta domain
Finite CP	TP	vP

- Crucially, each type may project larger, non-default structures, provided the semantic requirements (PRO and temporal interpretation) are not violated.

Background: PC vs EC

- The three semantic types form an implicative scale, known as the Implicative Complementation Hierarchy (ICH).

Most Independent		Least Independent
Least Transparent	Proposition >> Situation >> Event	Most Transparent
Least Integrated	(PC) (EC)	Most Integrated

adopted from Wurmbrand & Lohninger (2023:188)

- Mandarin is no exception to this cross-linguistic pattern (He, 2020, 2024; C.-T.J. Huang, 2022; Liu & Yip, 2026, among others).

Background: Overt Controlee

- The study of control theory in Mandarin is closely intertwined with the ‘finiteness debate’.
    - Mandarin lacks overt finiteness morphology, which makes other syntactic diagnostics essential.

Background: Overt Controlee

- The study of control theory in Mandarin is closely intertwined with the ‘finiteness debate’.
  - Mandarin lacks overt finiteness morphology, which makes other syntactic diagnostics essential.

- The availability of overt controlees in embedded clauses has served as a central diagnostic for clausal finiteness in Mandarin.

- Judgments on the availability of overt controlees vary substantially across researchers and constructions (Huang, 1982; Hu et al., 2001; Grano, 2015; Zhang, 2016; N.Huang, 2024; Yan, 2022; D.Li, 2025; and others).

Background: Overt Controlee

- A distinction in overt controlee licensing between PC and EC verbs ?

- D. Li (2025:347) claims that "only PC verbs license overt controlees in Mandarin":
  - PC: ✓ 李四 打算 [明年 夏天 (他) 去 旅游] (34.a)
    Lisi plan next-year summer (he) go travel
    "Lisi plans to go travel next summer".
  - EC: ✗ 李四 设法 [明年 夏天 (*他) 去 旅游] (34.b)
    Lisi try next-year summer (*he) go travel
    "Lisi tries to go travel next summer. "

- Yan (2022:33) similarly reports that Mandarin PC and EC differ in their tolerance of overt controlees:
  - PC: ✓ 张三 [打算 (他) 六点 出发] (52.a)
    Zhangsan plan he six-o'clock leave
    "Zhangsan plans to leave at 6 o'clock."
  - EC: ✗ 张三 [设法 (*他) 六点 出发] (52.b)
    Zhangsan try he six-o'clock leave
    "Zhangsan tries to leave at 6 o'clock. "

Research Questions

- Do PC verbs really allow overt controlees?
- Do EC verbs have extra penalty for overt controlees, compared to PC?
- If EC verbs do have extra penalty to PC verbs, is it conditioned by the C-selection difference?

Experiment 1: Design

- Crossing two factors: **Controlee Overtness** (*Null* v.s. *Overt*) × **Verb Type** (*Belief verbs* v.s. *PC* v.s. *EC*)
  - Belief verbs | Null -- Belief verbs | Overt
  - PC | Null -- PC | Overt
  - EC | Null -- EC | Overt

Belief verbs	觉得 (‘think’)	认为 (‘think’)	确认(‘confirm’)	声称(‘claim’)
PC	打算 (‘plan’)	计划 (‘plan’)	准备(‘prepare’),	争取(‘endeavour to’)
EC	尝试 (‘try’)	设法 (‘try’)	能够 (‘be able to’)	试图 (‘try’)

- 6 conditions × 4 items + 48 fillers = 72 sentences/list
- 48 native speaking participants, recruited from Prolific.
- Acceptability judgments: rating on a scale of 1 ~ 7 (1= completely ungrammatical, 7 = completely grammatical)

Experiment 1: Design

- Manner adverbs (vP internal) are added to the complement clauses.

- Sample items for experiment 1.

这个 小偷 **认为** [(他) 可以 悄悄 偷走 那枚 钻石] (Null/Overt | Belief)
this.CL thief think he can secretly steal that.CL diamond
"This thief thinks that he can secretly steal that diamond."

这个 小偷 **打算** [(他) 悄悄 偷走 那枚 钻石] (Null/Overt | Partial)
this.CL thief plans he secretly steal that.CL diamond
"This thief plans to secretly steal that diamond."

这个 小偷 **尝试** [(他) 悄悄 偷走 那枚 钻石] (Null/Overt | Exhaustive)
this.CL thief try he secretly steal that.CL diamond
"This thief tries to secretly steal that diamond."

Experiment 1: Logic & Predictions

Three primary working hypotheses

- By default, PC verbs select TP complements; EC verbs select vP complements.
- Only finite TPs can license overt subjects.
- Overy subjects occupy [Spec TP].

- *Without overt controlees*, all three verb types (Belief, PC, EC) should be fine:
  - The manner adverbs do not interfere with the semantic requirements of PC and EC verbs.
  - The manner adverbs are vP internal, so they respect PC and EC's default C-selections:

Verb types	*Complements*
Belief verbs	$[_{CP}$ … $[_{TP}$ ***pro*** $[_{vP}$ manner adverb $[_{vP}$ <***pro***> […]]]]]
PC verbs	$[_{TP}$ … $[_{vP}$ manner adverb $[_{vP}$ **PRO** […]]]]
EC verbs	$[_{vP}$ manner adverb $[_{vP}$ **PRO** […]]]

Ex.1 Verbs and complements, Null Controlee

Experiment 1: Logic & Predictions

Three primary working hypotheses

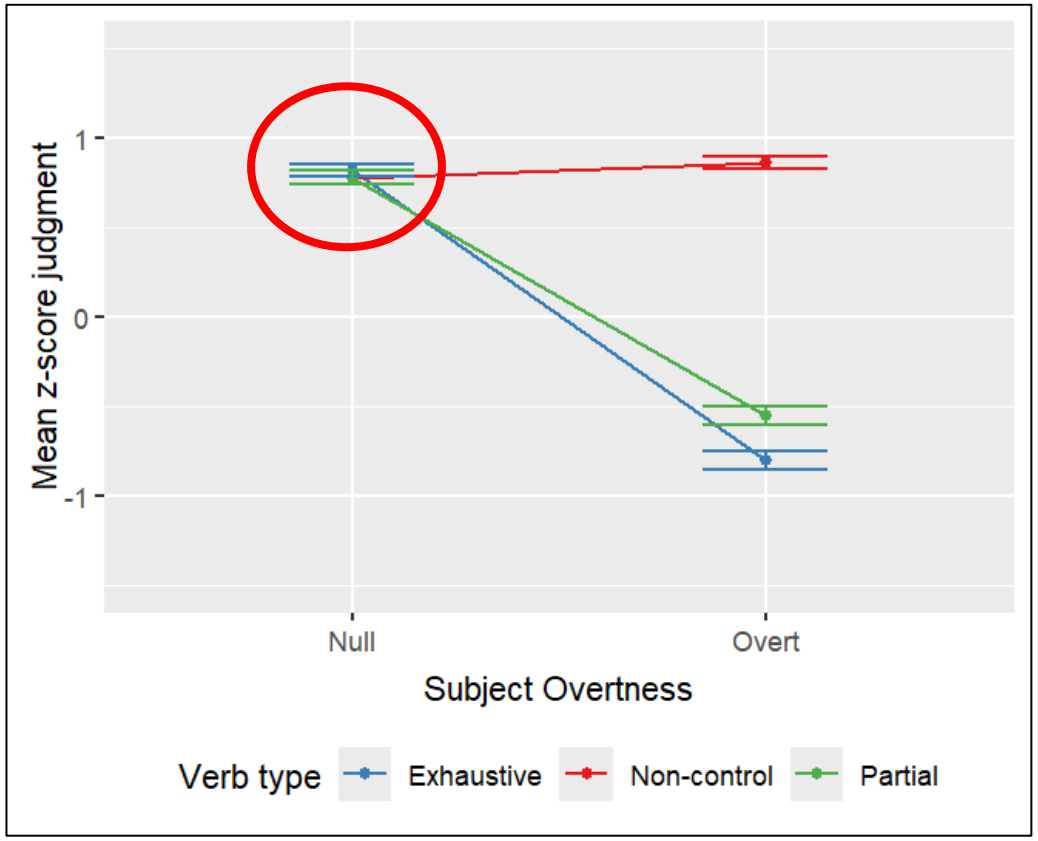
- By default, PC verbs select TP complements; EC verbs select vP complements.
- Only finite TPs can license overt subjects.
- Overy subjects occupy [Spec TP].

- *After adding overt controlees*:
  - **BV penalty** = ∅
    Belief verbs remain to be acceptable.
  - **PC penalty** = **finiteness violation**
    If PC complements are indeed nonfinite, overt controlees are not licensed.
  - **EC penalty** = **finiteness violation** + **non-default C-selection**
    (1) overt controlees are not licensed;
    (2) the extra cost for a non-default complement size — since EC verbs prefer *v*P but an overt subject requires IP/TP. [1]

[1] Less preferred subcategorization increases processing cost (Trueswell et al., 1993; Garnsey et al., 1997)

Experiment 1: Results

‘The three-way split’

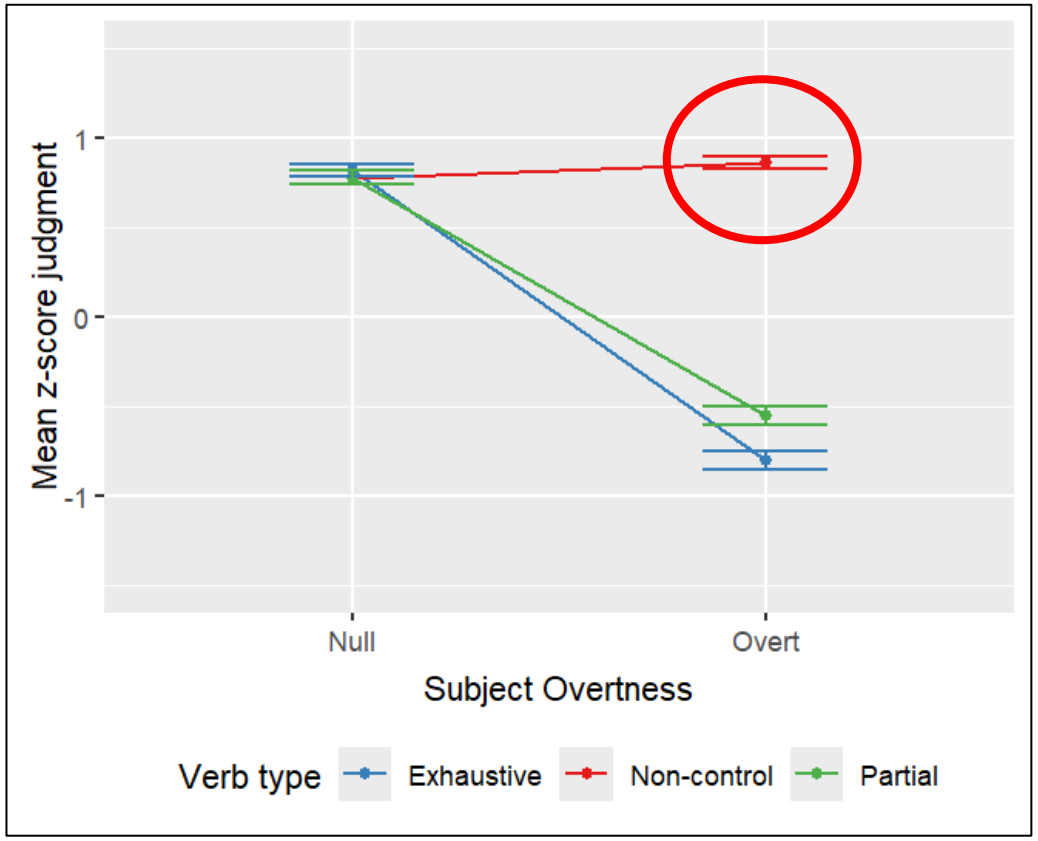


Ex.1 Interaction plot

- With null controlees, all three verb types are acceptable.

Experiment 1: Results

‘The three-way split’

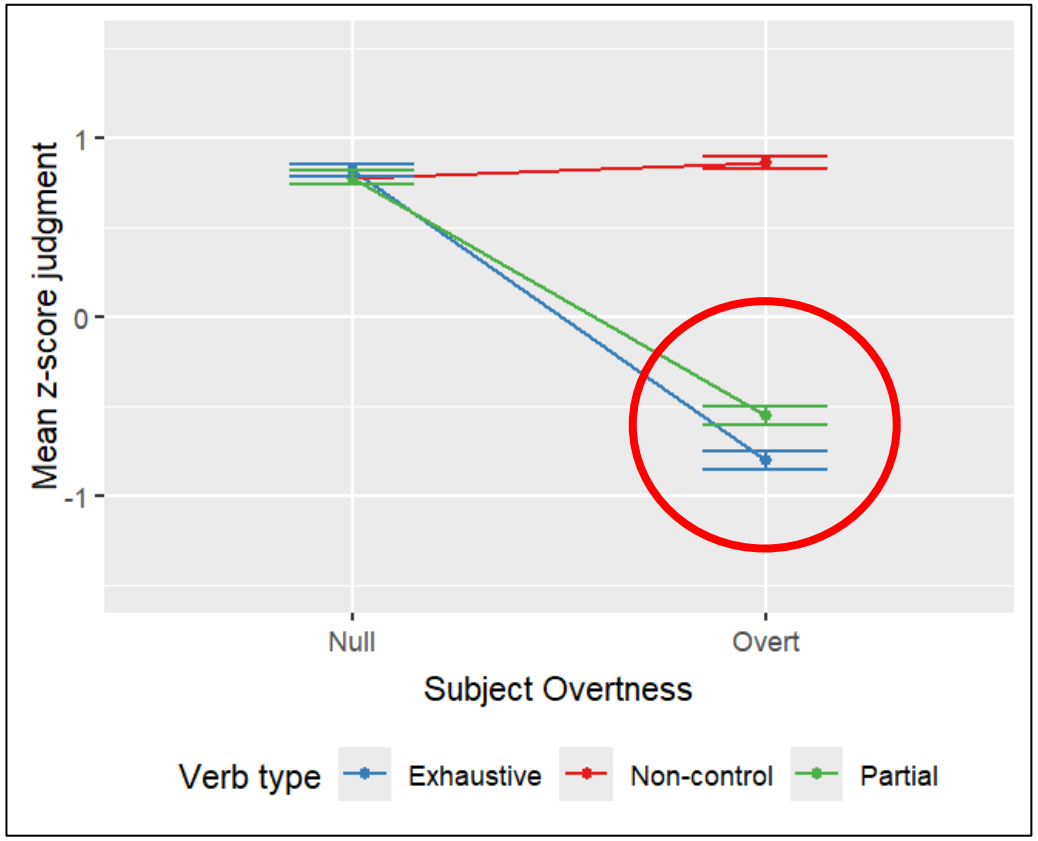


Ex.1 Interaction plot

- With null controlees, all three verb types are acceptable.
- Belief verbs with overt controlees remain to be acceptable, as predicted.

Experiment 1: Results

‘The three-way split’

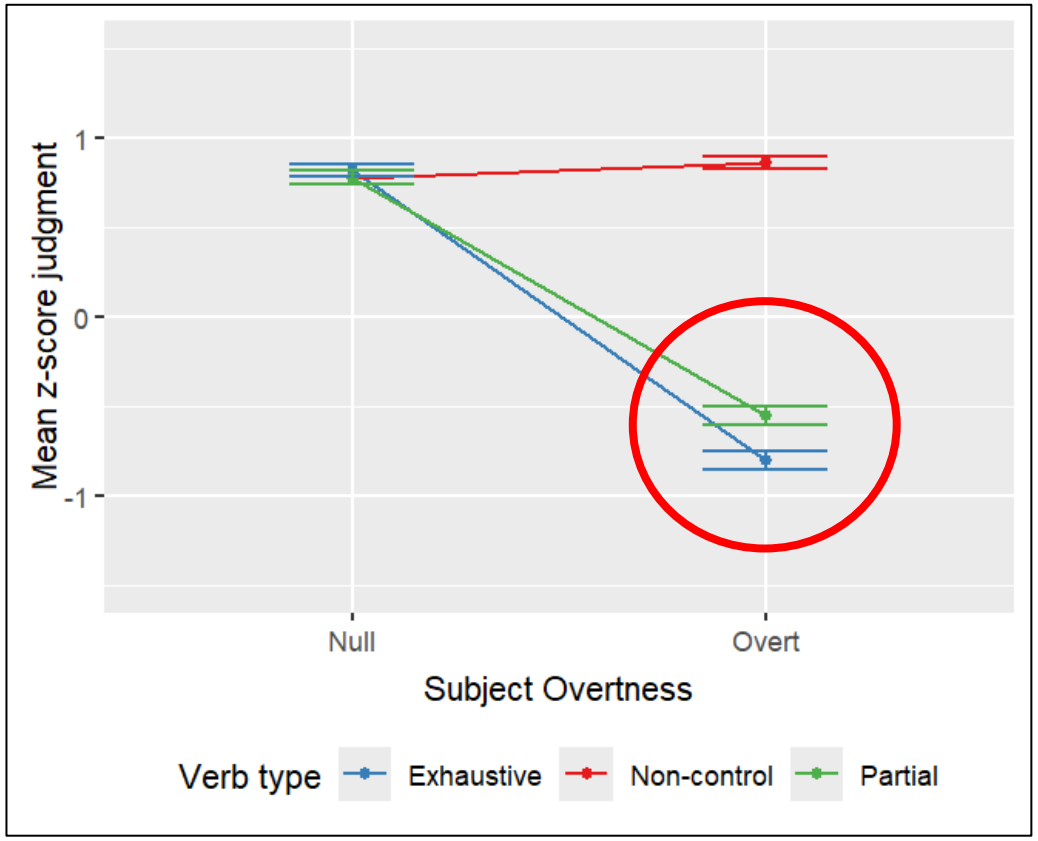


Ex.1 Interaction plot

- With null controlees, all three verb types are acceptable.
- Belief verbs with overt controlees remain to be acceptable, as predicted.
- PC + Overt controlee = Degradation ($p < .0001$).
- EC + Overt controlee is even worse.
- The PC-EC difference after adding overt controlees is statistically significant ($p < .005$).

Experiment 1: Results

‘The three-way split’



Ex.1 Interaction plot

- With null controlees, all three verb types are acceptable.
- Belief verbs with overt controlees remain to be acceptable, as predicted.
- PC + Overt controlee = Degradation ($p < .0001$).
- EC + Overt controlee is even worse.
- The PC-EC difference after adding overt controlees is statistically significant ($p < .005$).
- A ‘three-way split’ of acceptability with overt controlees.

Experiment 1: Results

- PC predicates do not allow overt controlees, contra D.Li (2025) and Yan (2022).
- However, both are correct in observing that EC verbs are extra bad compared to PC.
- A three-way split is therefore observed after adding overt controlees: $Belief\ verbs \gg PC > EC$

Experiment 2: Logic & Predictions

Experiment 2: Logic & Predictions

- Ex.1 has observed extra penalty for EC after adding overt controlees.
- The extra penalty is attributed to default complement size differences between PC and EC.
- If this is on the right track, then if we force all conditions to involve larger complement sizes, no extra penalty for EC should be observed after adding overt controlees.

Experiment 2: Logic & Predictions

- Everything the same, but the manner adverbs are replaced by future-reference temporal adverbs (assumed to be structurally higher).

- Sample items for experiment 2:

这个 小偷 认为 [明天晚上 (他) 可以 偷走 那枚 钻石] (Null/Overt | Belief)
this.CL thief thinks tomorrow night he can steal that.CL diamond
"This thief thinks that he can steal that diamond tomorrow night."

这个 小偷 打算 [明天晚上 (他) 偷走 那枚 钻石] (Null/Overt | Partial)
this.CL thief plans tomorrow night he steal that.CL diamond
"This thief plans to steal that diamond tomorrow night."

这个 小偷 尝试 [明天晚上 (他) 偷走 那枚 钻石] (Null/Overt | Exhaustive)
this.CL thief try tomorrow night he steal that.CL diamond
"This thief tries to steal that diamond tomorrow night."

Experiment 2: Logic & Predictions

- **Without overt controlees**

The temporal adverbs are structurally higher ---> EC forced to have larger complements from the onset:

Verb types	*Complements*
Belief verbs	$[_{CP}$ … $[_{TP}$ temporal adverb ***pro*** $[_{vP}$ <***pro***> […]]]]
PC verbs	$[_{TP}$ temporal adverb $[_{vP}$ **PRO** […]]]
EC verbs	$[_{TP/IP}$ temporal adverb $[_{vP}$ **PRO** […]]]

Ex.2 Verbs and complements, Null Controlee

Experiment 2: Logic & Predictions

- **Without overt controlees**
  - **BV penalty** = ∅
    Belief verbs are grammatical.
  - **PC penalty** = ∅
    PC verbs are grammatical.
  - **EC penalty** = **non-default C-selection** + **temporal simultaneity violation**
    (1) temporal adverbs signal larger than vP complements (non-default);
    (2) the future time reference violates the temporal-simultaneity constraint.

Experiment 2: Logic & Predictions

- **Without overt controlees**
  - **BV penalty** = ∅
    Belief verbs are grammatical.
  - **PC penalty** = ∅
    PC verbs are grammatical.
  - **EC penalty** = **non-default C-selection** + **temporal simultaneity violation**
    (1) temporal adverbs signal larger than vP complements (non-default);
    (2) the future time reference violates the temporal-simultaneity constraint.
- **After adding overt controlees**
  - **BV penalty** = ∅
    Belief verbs are grammatical.
  - **PC penalty** = **finiteness violation**
    Overt controlees are not licensed.
  - **EC penalty** = **non-default C-selection** + **temporal simultaneity violation** + **finiteness violation**
    (1) temporal adverbs signal larger than vP complements (non-default);
    (2) the future time reference violates the temporal-simultaneity constraint;
    (3) Overt controlees are not licensed.

Experiment 2: Logic & Predictions

- **Without overt controlees**
  - **BV penalty** = ∅
  - **PC penalty** = ∅
  - **EC penalty** = **non-default C-selection** + **temporal simultaneity violation**
- **After adding overt controlees**
  - **BV penalty** = ∅
  - **PC penalty** = **finiteness violation**
  - **EC penalty** = **non-default C-selection** + **temporal simultaneity violation** + **finiteness violation**

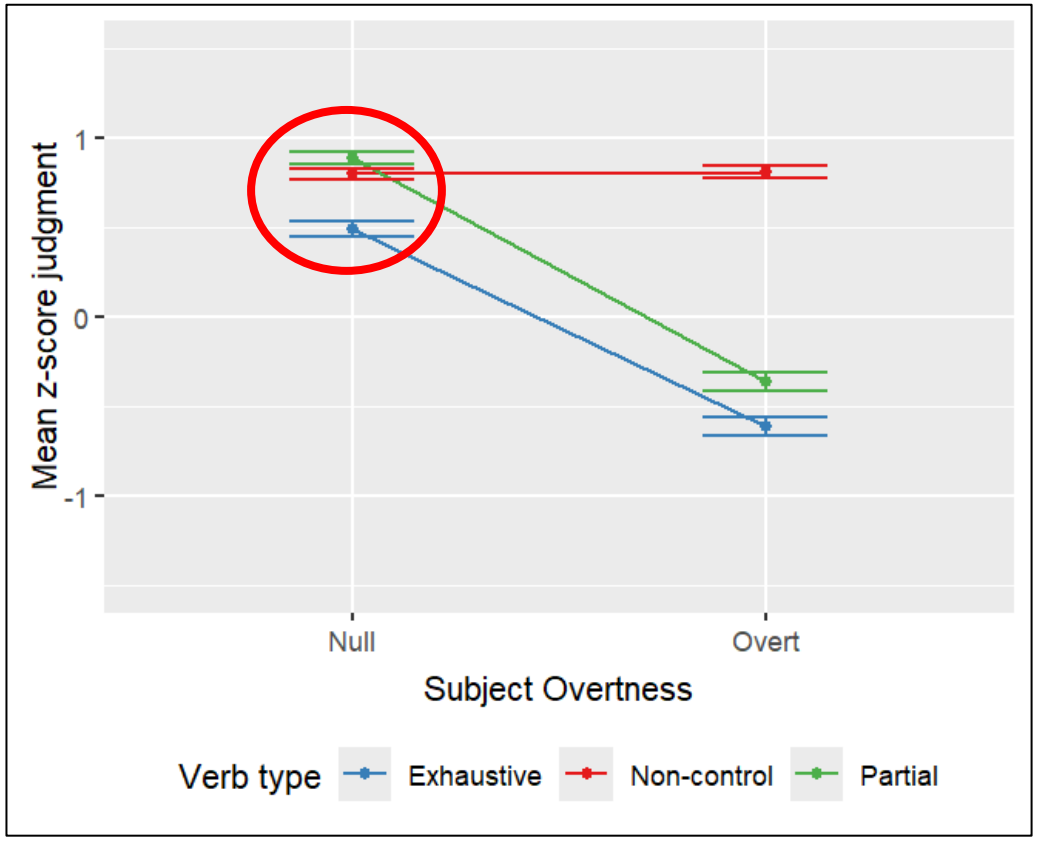
- Both PC and EC only have one source of penalty after adding overt controlees – finiteness violation.

  EC has already paid the 'C-selection cost' in its null controlee baseline.

  No extra penalty for EC.

Experiment 2: Results

‘No extra EC penalty’

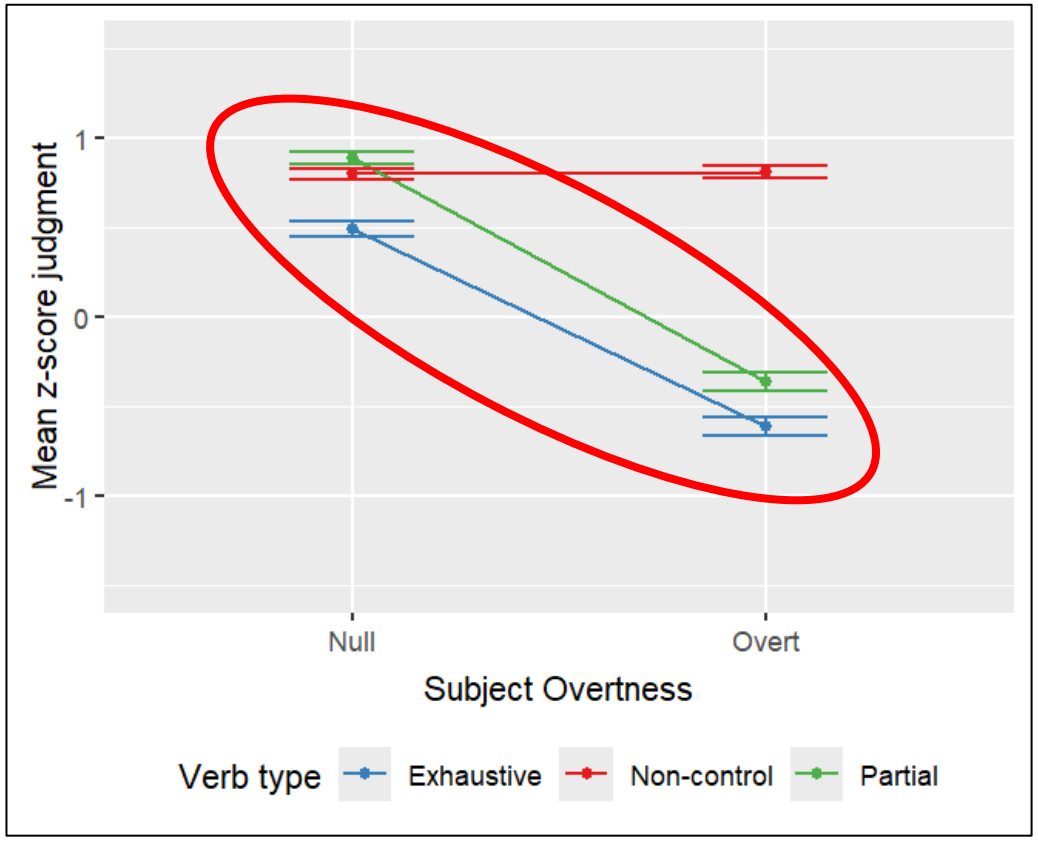


Ex.2 Interaction plot

- With null controlees, EC is already degraded as predicted.

Experiment 2: Results

‘No extra EC penalty’



Ex.2 Interaction plot

- With null controlees, EC is already degraded as predicted.
- After introducing overt controlees, both PC and EC dropped significantly.
- PC and EC lines are parallel – no extra penalty for EC ($p = 0.13$).

Experiment 2: Results

- The extra penalty of EC overt controlees compared to PC is conditioned by complement sizes.
- If an EC verb already has a non-default C-selection before adding overt controlees, no extra penalty would be induced.
- The observed difference in tolerance of EC and PC, as reported by D.Li (2025) and Yan (2022), is better analyzed as a side effect of C-selection difference.

Answering the Research Questions

- PC verbs disallow overt controlees.
- EC verbs disallow overt controlees, and can incur extra penalties relative to PC verbs.
- The extra penalty of EC is conditioned by complement sizes.

References

Garnsy et al. (1997)

Grano, T. (2015)

Landau, I. (2001)

Li, D. (2024)

Liu, Y., & Yip, K. (2026)

Huang, C.-T.J. (1982)

Huang, C.-T.J. (2022)

He, Y. (2020)

He, Y. (2024)

Trueswell et al. (1993)

Wurmbrand, S., & Lohninger, D. (2023)

Huang, N. (2018)

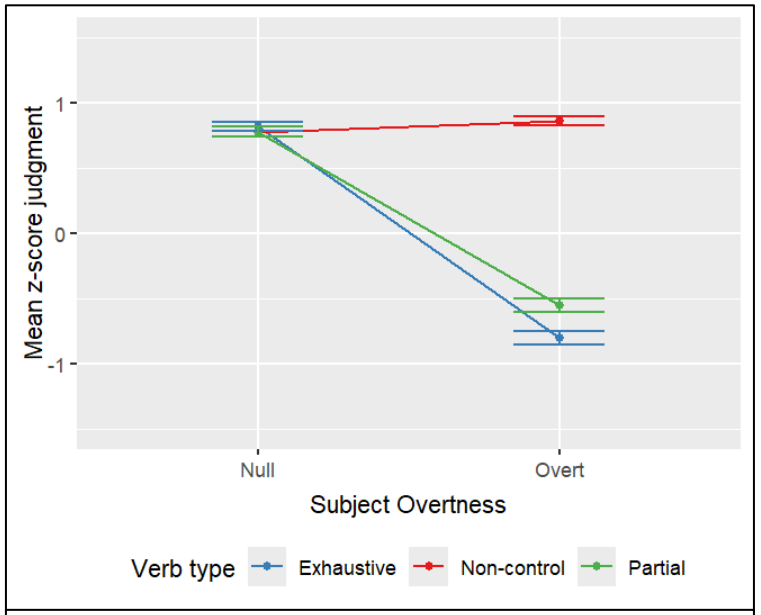
Huang, N. (2024)

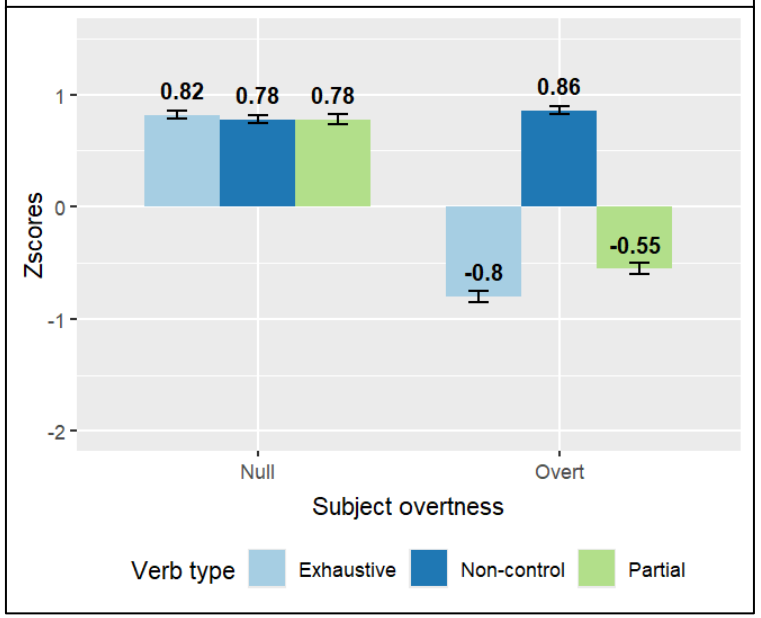
Hu, J., Pan, H., (2001)

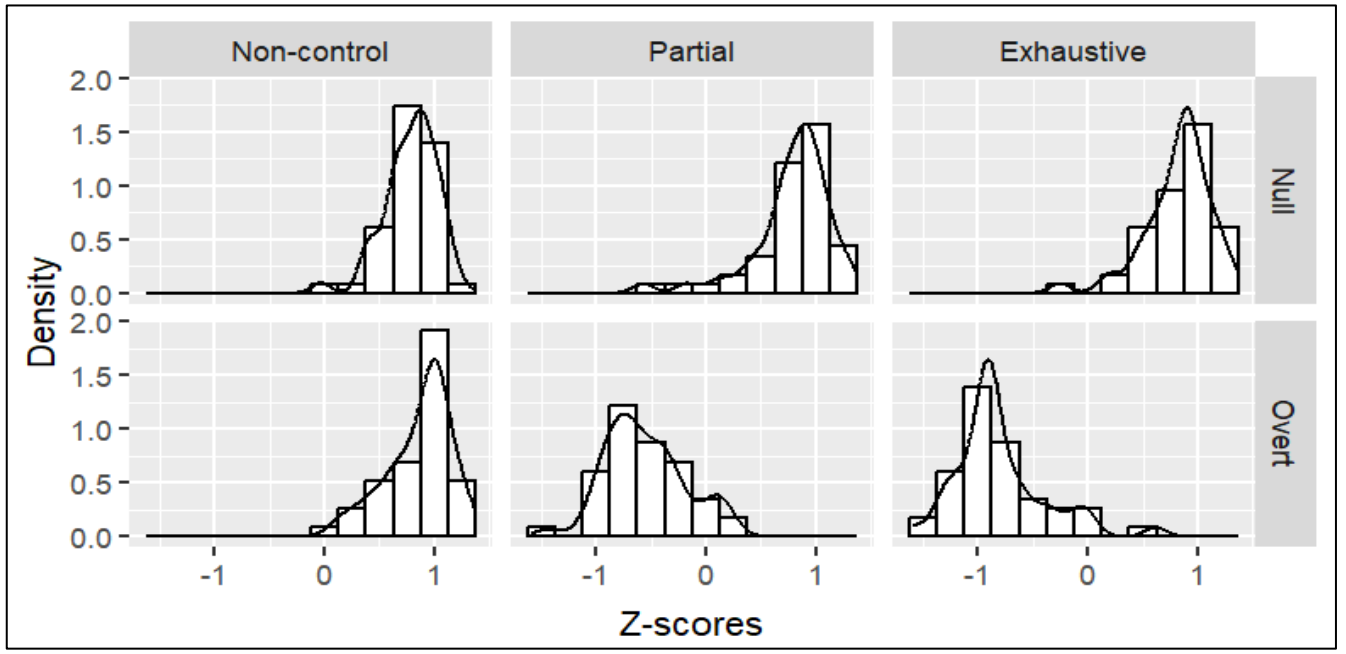
Yan, L. (2022)

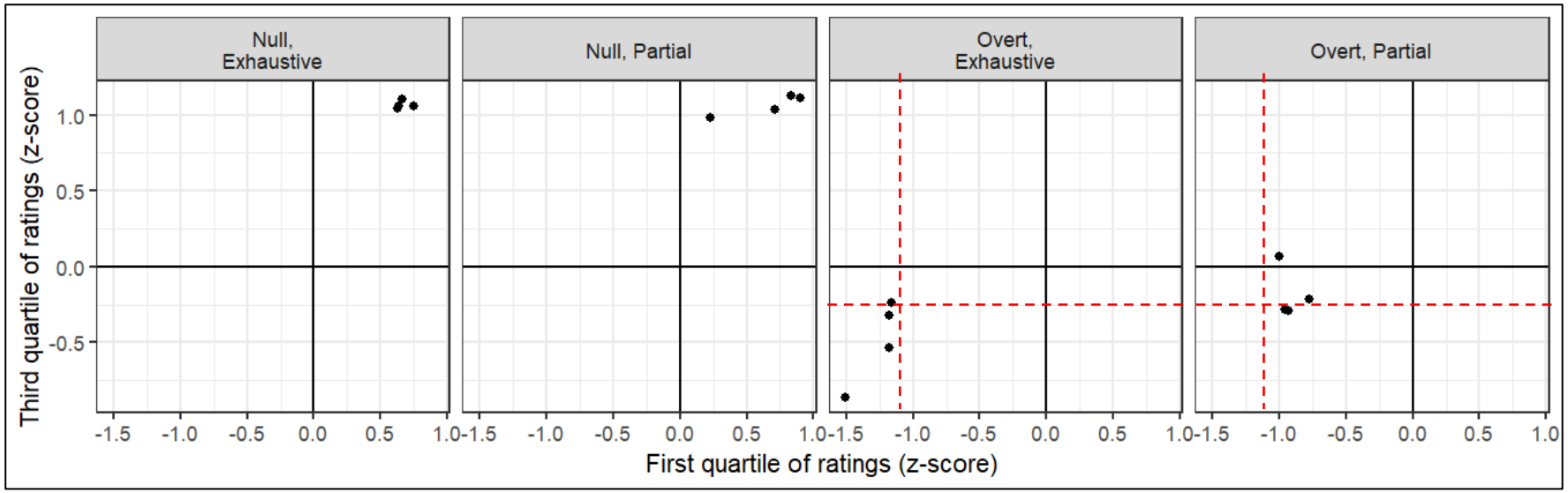
Zhang, N. (2016)

Appendix: Experiment 1









Appendix: Experiment 2

